Practice Quiz 11B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 11

Compare the integers with an arithmetic progression.

1. Let $P = \{5k + 2 : k \in \mathbf{Z}\}$. Define a function $g : \mathbf{Z} \to P$ and prove that it is a bijection. Conclude that P and $\mathbf{Z}$ are equinumerous.

   Solution Define $g(k) = 5k + 2$ for $k \in \mathbf{Z}$. This is an element of P. If $g(j) = g(k)$, then $5j+2 = 5k+2$, hence $j = k$; so g is one-to-one. If $p \in P$, then $p = 5k+2 = g(k)$ for some integer k, so g is onto P. Therefore g is a bijection, and $P \sim \mathbf{Z}$. ∎

Solutions prepared using GPT-6 in the Codex harness.